



JOINT QUANTUM INSTITUTE
SCHINE GROUP

MOT Coil Current Switching Controller Manual

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Quick Reference Specifications

Max Current Output	20 A
Max MOT Coil Supply Input	15 VDC
Set Point Ratio	-0.501 V/A
Monitor Ratio	-0.101 V/A

Set point	Rise Time (1/e)	Fall Time (1/e)
-2 V	3.00 ms	2.40 ms
-4 V	5.50 ms	3.40 ms
-6 V	8.90 ms	4.50 ms
-8 V	15.0 ms	6.20 ms
-10 V	22.5 ms	6.50 ms

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1 Introduction

The MOT Coil Controller (MCC) is an experimental apparatus designed to drive a pair of Helmholtz coils and, at maximum operational load, switch the direction of current in one of the coils. The system employs a high-current capable H-bridge that can handle switching up to 20A with an integrated protection system that prevents switching until the current has decreased below a threshold value. The operational status of the control box is indicated by two LEDs.

The box operates in one of two distinct states: Helmholtz or Anti-Helmholtz configuration, referred to as State Green and State Red for simplicity.

The H-bridge design offers versatility beyond its primary application, making it suitable for other high-current switching applications where switching times are not required to be less than 30 ms and protection against back-EMF is important.

2 System Overview

2.1 Problem Encountered

2.1.1 H-bridge + Relay board combination

The MOSFET utilized in the H-bridge contain inherent parasitic diodes, all oriented in the same direction as can be seen in Figure 1. This configuration creates challenges for rapid current dissipation when switching between operational states. The problem became apparent only after the circuit had already been assembled.

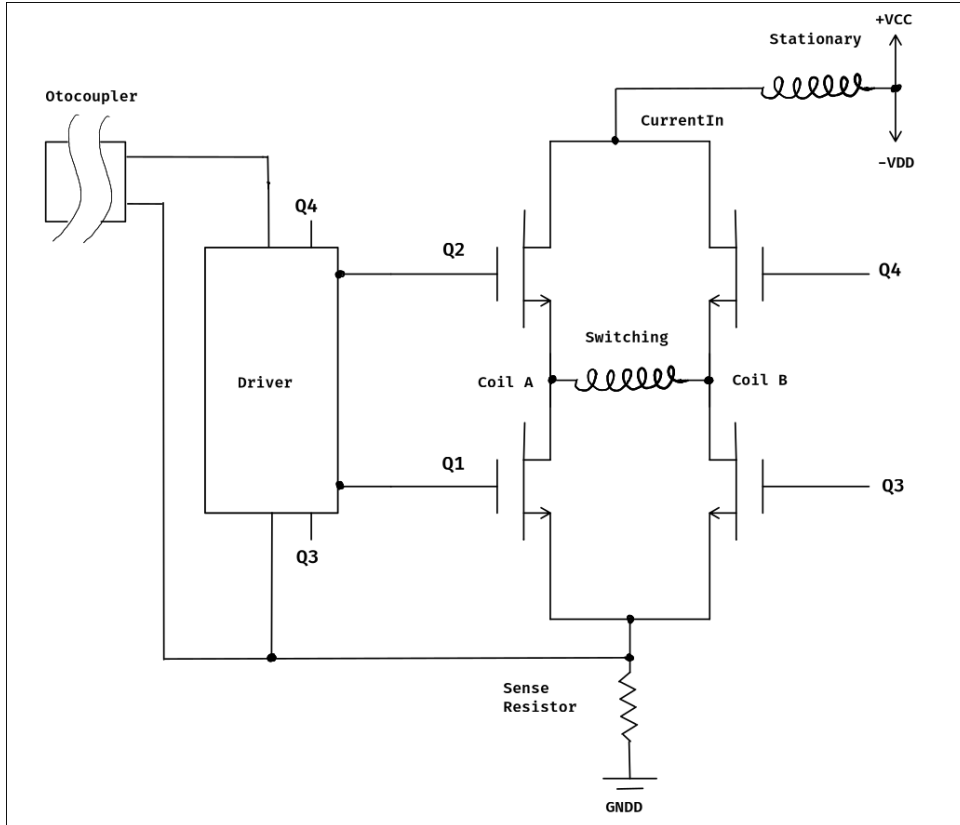


Figure 1: MOSFET Arrangement with Parasitic Diodes

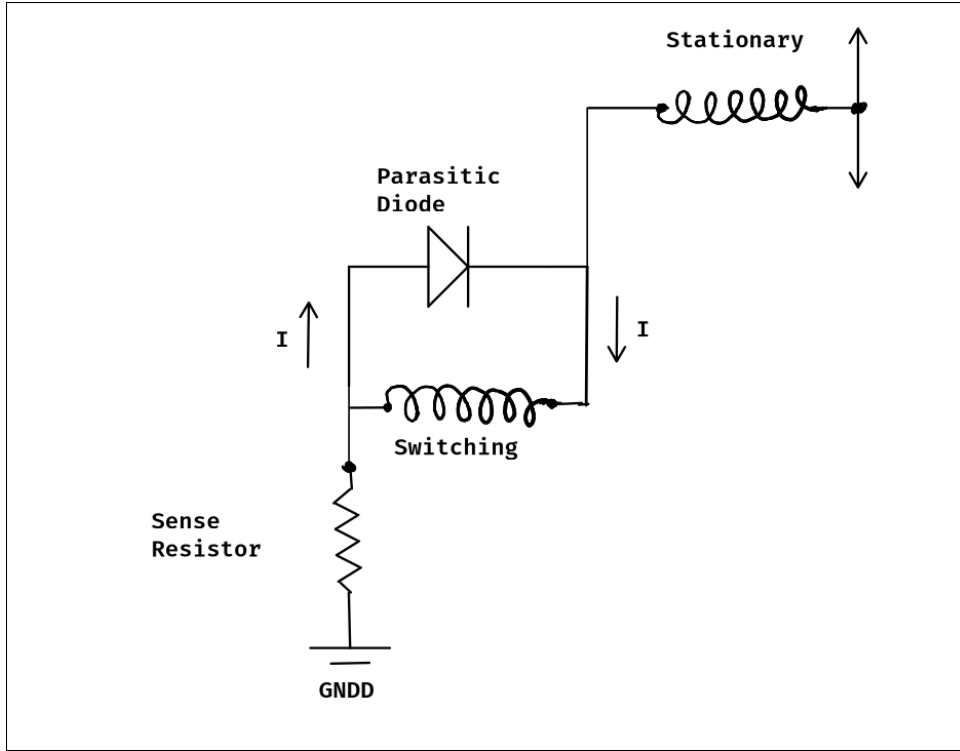


Figure 2: Simplified Circuit Showing Current Trapping Effect

Instead of allowing current to dissipate quickly during state transitions (Green to Red or Red to Green), these diodes create a closed loop with the switching coil, effectively trapping the dissipating current and resulting in extended rise and fall times as shown in Figure 2. To address this limitation, a separate relay board was incorporated into the design. This board connects to only one of the four MOSFETs, which now functions as a switch to toggle the relays while maintaining the protection circuit functionality on the H-bridge.

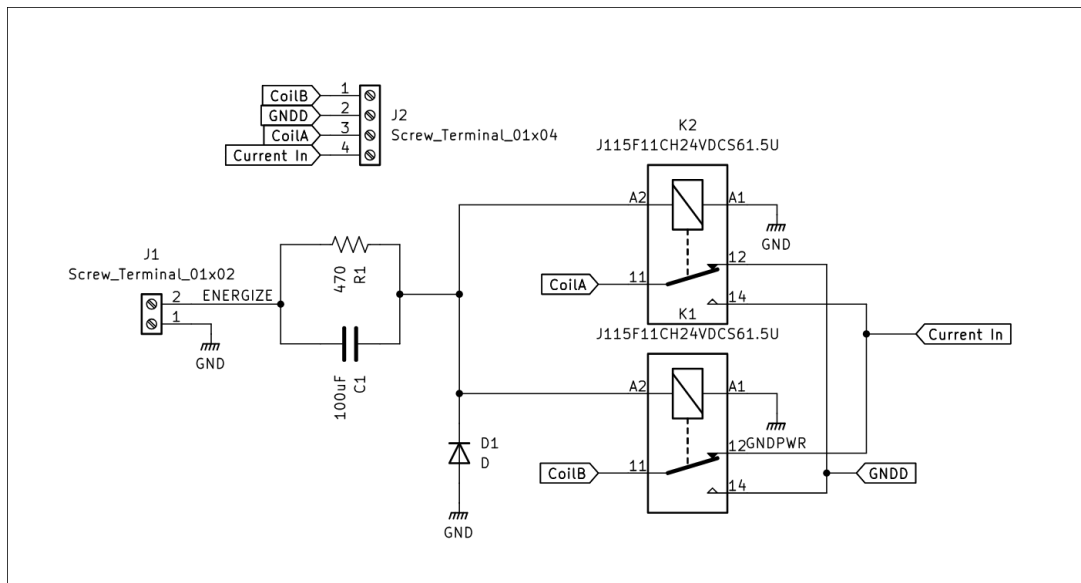


Figure 3: Relay Board Circuit

2.1.2 DC-DC converted Noise on H-bridge

The protection circuit initially exhibited noticeable noise originating from the DC-DC converter, which was visible during oscilloscope measurements. This issue was mitigated by later soldering a large capacitor between the converter and ground. This practice is commonly used in commercial isolated power supplies where the component is referred to as class-Y capacitor. While some noise remains present, it has been significantly reduced to acceptable levels. In the event of noise-related issues, it is recommended to inspect the capacitor connections to ensure proper functionality.

2.1.3 Modified FET Board vs Bipolar Configurations

An attempt was made to modify the FET board from the BIAS Coil Controller to increase the maximum supply coil voltage beyond 15V by turning the bipolar board into a Mono-Polar design. However, removing the bipolar nature of the original FET board resulted in the loss of its push-pull characteristics. Additionally, difficulties in tuning the PID loop with the new MOSFETs led to significantly increased switching times. Consequently, the decision was made to retain the original bipolar FET board configuration. For additional details regarding the bipolar FET board and PID tuning procedures, please consult the BIAS Coil Controller manual.

2.2 Final configuration

The final MOT Coil Current Controller configuration consists of three primary components working in tandem: the Control Board, MOSFET Board, and the H-Bridge plus Relay Boards. These components are housed in a 3U-D2 Aluminum mount Rack with front and rear panel interfaces as described in Section 4.

As shown in Figure 4, the internal layout has been derived from the BIAS Coil Controller which provides the primary current control functionality, while the H-Bridge circuit (detailed in Section 5.2 and 5.3) details the Protection against Back-EMF. The addition of the Relay board enables current direction switching while resolving the current trapping issues described in Section 2.1.1.

In this configuration, the system operates in one of two distinct states:

- **State Green (Helmholtz Configuration):** Both coils have current flowing in the same direction, creating a uniform magnetic field in the center region.
- **State Red (Anti-Helmholtz Configuration):** Current flows in opposite directions through the two coils, creating a magnetic field gradient with a zero-crossing at the center point.

The transition between these states is managed by the protection circuit (detailed in Section 5.3), which ensures switching only occurs when the current has decreased below a safe threshold. This threshold can be adjusted using the potentiometer indicated in Figure 4.

The final design accommodates the challenges outlined in Section 2.1, achieving reliable switching operation while maintaining safety against potential back-EMF damage. However, It is important to note that due to difficulties with designing a Mono-polar FET board, the MCC is not quite able to reach the targeted 20A.

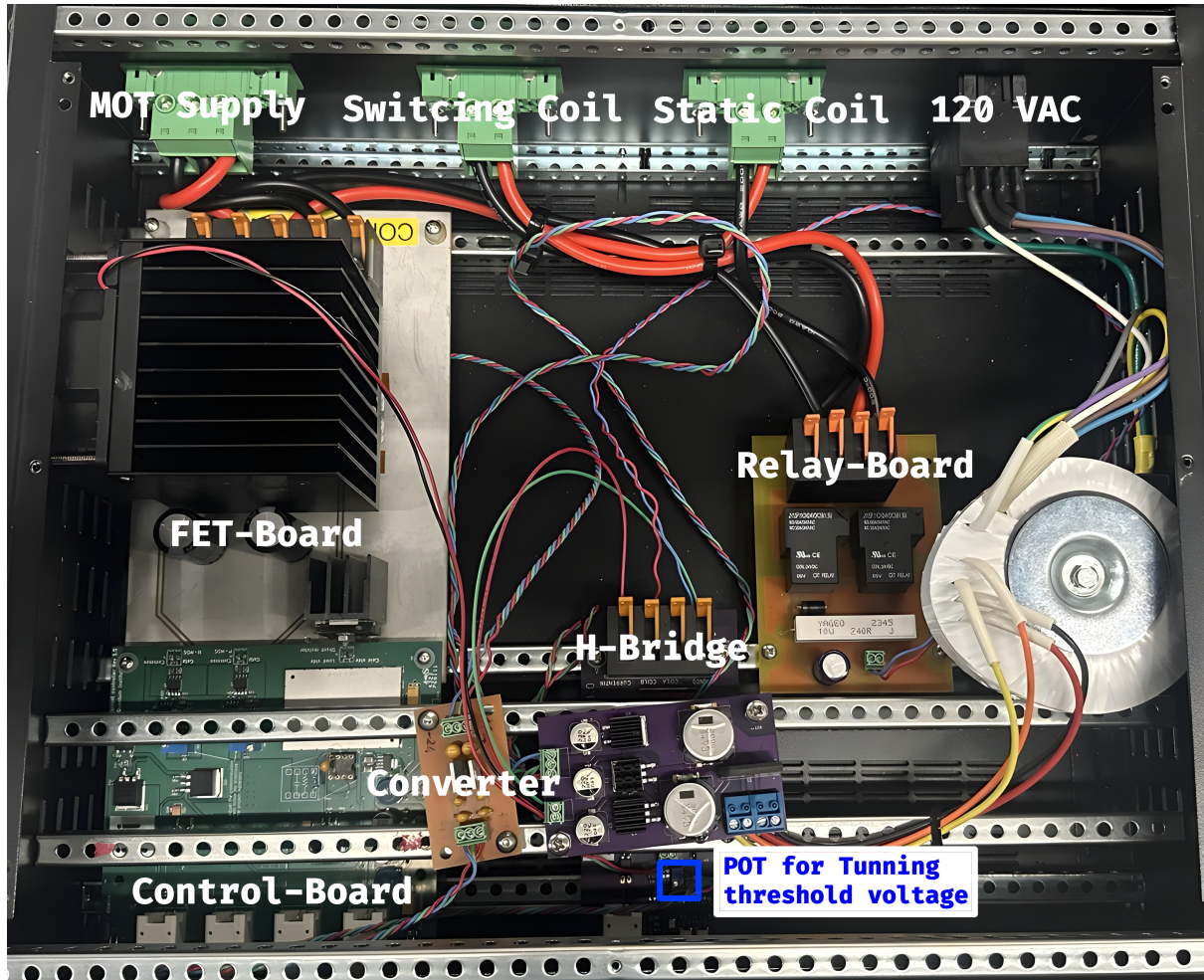


Figure 4: Top View of MCC Box Interior

3 Specifications

3.1 Electrical Specifications

Please refer to the Quick Reference table for conversion ratios. Operating Voltages are provided in Table 1 below.

3.2 Relay Specifications

The relays are mechanical devices and will likely need replacement after a couple of years of regular use. Due to their mechanical nature, it is critical not to drive the relays at high frequency as this will significantly reduce their operational lifespan. See Table 2 for additional details.

3.3 Response Specifications

Please refer to the Quick Reference table for the $1/e$ value of the Rise and Fall times. See Table 3 for Full oscilloscope captures of the Rise and Fall waves. The yellow trace is the voltage across the switching coil, while the blue trace is the set point.

Parameter	Specification	Notes
MOT Coil Power Supply Input	± 15 V	Must not exceed 15 V
Maximum Current Output	18-19 A	Cannot quite reach the expected 20 A the MCC was designed for
Main Input Power	120 VAC	Standard wall outlet power
Control Board Operating Voltage	± 24 V	Supplied by 120 VAC to 48 VAC transformer
H-Bridge Operating Voltage	± 15 V	Supplied by the Converter board from ± 24 V

Table 1: Electrical Specifications

Parameter	Specification
Relay Type	CIT Relay and Switch J115F Series
Contact Form	SPDT (1 Form C)
Contact Rating	50 A
Switching Voltage	277VAC, 110VDC (Maximum)
Coil Voltage	24VDC
Coil Current	62.5 mA
Coil Resistance	384 Ohms
Operate Time	15 ms
Release Time	10 ms
Operating Temperature	-55°C ~ 125°C
Contact Material	Silver Tin Oxide Indium Oxide (AgSnOInO)

Table 2: Relay Specifications

4 Rear and Front Panel Connections

4.1 Front Panel

The front panel features three input connections (Set Point, Disable, and H-Bridge TTL), two output connections (Monitor and Error Signal), and four indicator LEDs for system status.

4.1.1 Input Connections

- **Set Point:** The voltage input port for setting the desired current through the MOT coil output on rear panel.
- **Disable:** Allows the user to immediately disable operation of the channel by sending a TTL signal. The channel is disabled while the signal is high and resumes normal operation once the signal is low. The ‘DISABLED’ LED lights up while the channel is inoperable.
- **H-Bridge TTL:** Controls the operational state of the MCC through a 0-5V TTL signal. The left Protection LED indicates the state of the TTL signal: when the channel is low, the system is in State Green; when the channel is high, the system is

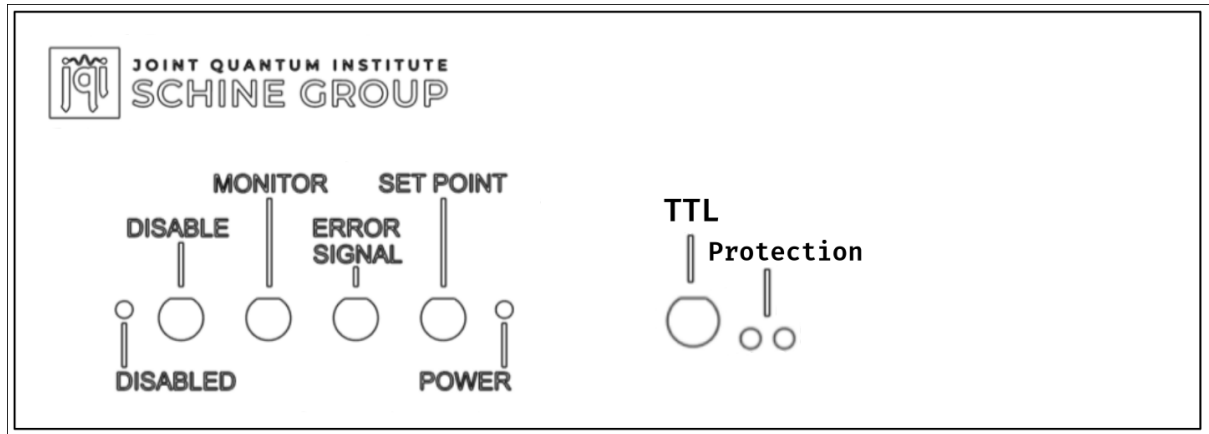


Figure 5: Front Panel Layout

in State Red. Note that the MCC will not switch states immediately if the current through the coils is not zero.

4.1.2 Output Connections

- **Monitor:** Outputs a voltage corresponding to the current flowing through the channel's output port on the rear panel.
- **Error Signal:** Outputs a voltage corresponding to the error in current between the current corresponding to the input voltage on 'SET POINT' and the 'MONITOR'.

4.2 Rear Panel

The rear panel features the two MOT Coils outputs and power inputs for the internal circuitry and MOT coils power supply.

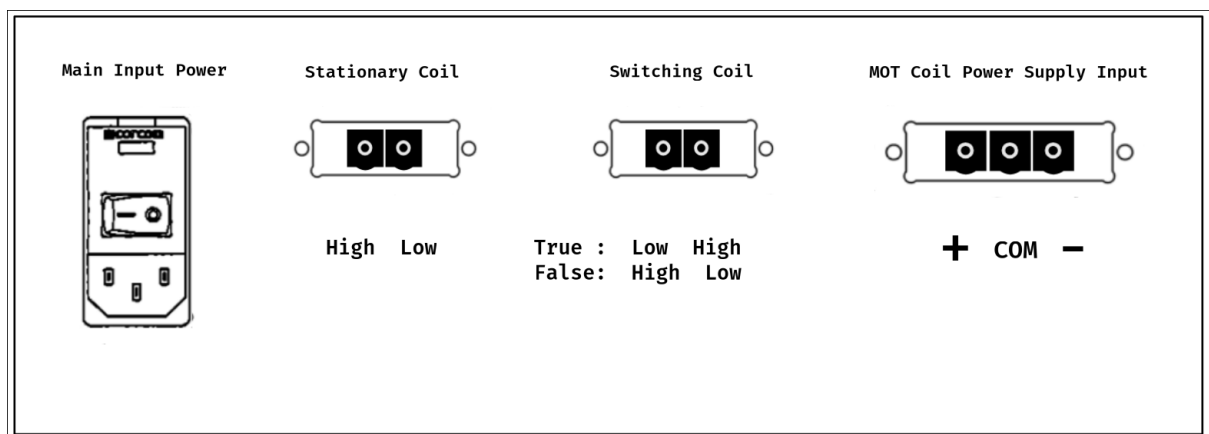


Figure 6: Rear Panel Layout

4.2.1 Input Connections

- **Main Power Input:** This input takes 120 VAC only. A rocker switch allows users to control power to the box while plugged in to the wall. A fuse holder limits the maximum current allowed into the box.
- **MOT Coil Power Supply Input:** The three-socket connector receives current from an external power supply to drive the MOT coils.

4.2.2 Output Connections

- **MOT Coil Output:** The two-socket connector provides power to the pair of coils. The left connector is designated for the switching coil. The right connector is used by the stationary coil. To switch the default state of the MCC, the switching coil ends can be swapped.

5 Circuit Design

5.1 Current Source Controller

For detailed specifications regarding the Current Source Controller, please refer to the BIAS Coil Controller manual. This section focuses on the connections and integration between the FET Board and the additional H-bridge and Relay Board components that constitute the MOT Coil Current Controller.

Figure 7 Shows the connections that have been established between the FET Board and the H-bridge and Relay Board.

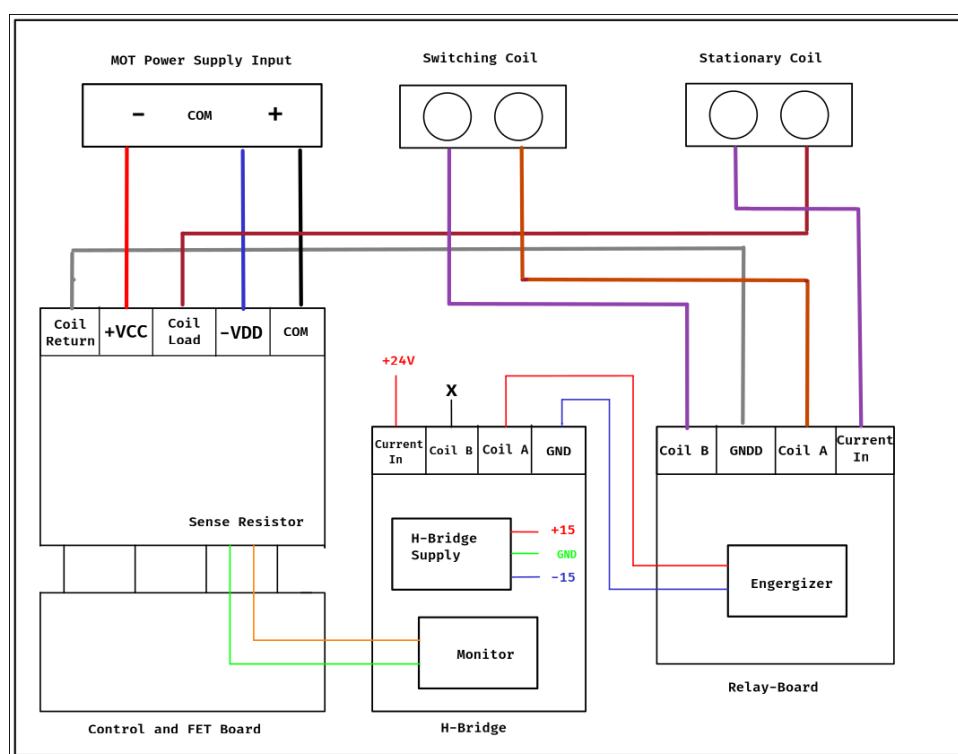


Figure 7: Block Diagram of FET Board, H-Bridge, and Relay Board Connections

5.2 H-Bridge Circuit

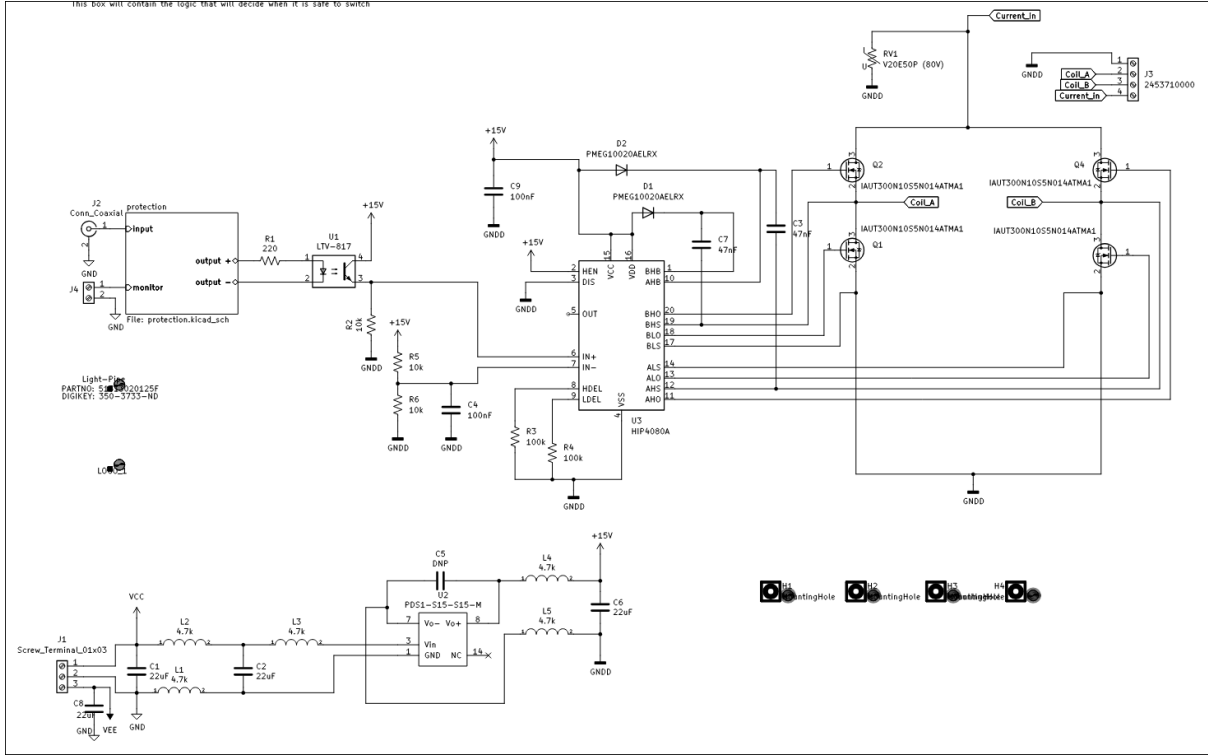


Figure 8: H-Bridge Circuit Schematic

The H-bridge circuit employs a simple design optimized for high-current switching applications. A key feature of this design is the optocoupler, which electrically isolates the control signals from the power circuit, allowing the H-bridge to operate as a floating circuit.

The HIP4080A driver IC serves as the core component that drives control signals to the MOSFETs. This commercially available specialized driver is designed specifically for high-current bridge applications and little modification was required to integrate into the H-bridge design. The power stage consists of four N-channel MOSFETs capable of 300A arranged in a full H-bridge configuration.

It is important to note that a large capacitor was later soldered between the DC-DC converter and ground to mitigate noise issues, but this modification is not reflected in the current schematic.

The circuit includes four primary connections: GND, COIL A, COIL B (representing the two ends of the switching coil), and CurrentIn. In the final configuration, CurrentIn is supplied with +24V. The default state of the MCC can be changed by connecting either COIL A or COIL B to the relay's energize port.

In the final design, the GND, COIL A, COIL B, and CurrentIn connections have been transferred to the relay board, with the GND terminal on the H-bridge as shown in Figure 7 is connected to the Main power supply ground, while the GND port on the relay is connected to the COM of the MOT Coil Power Supply.

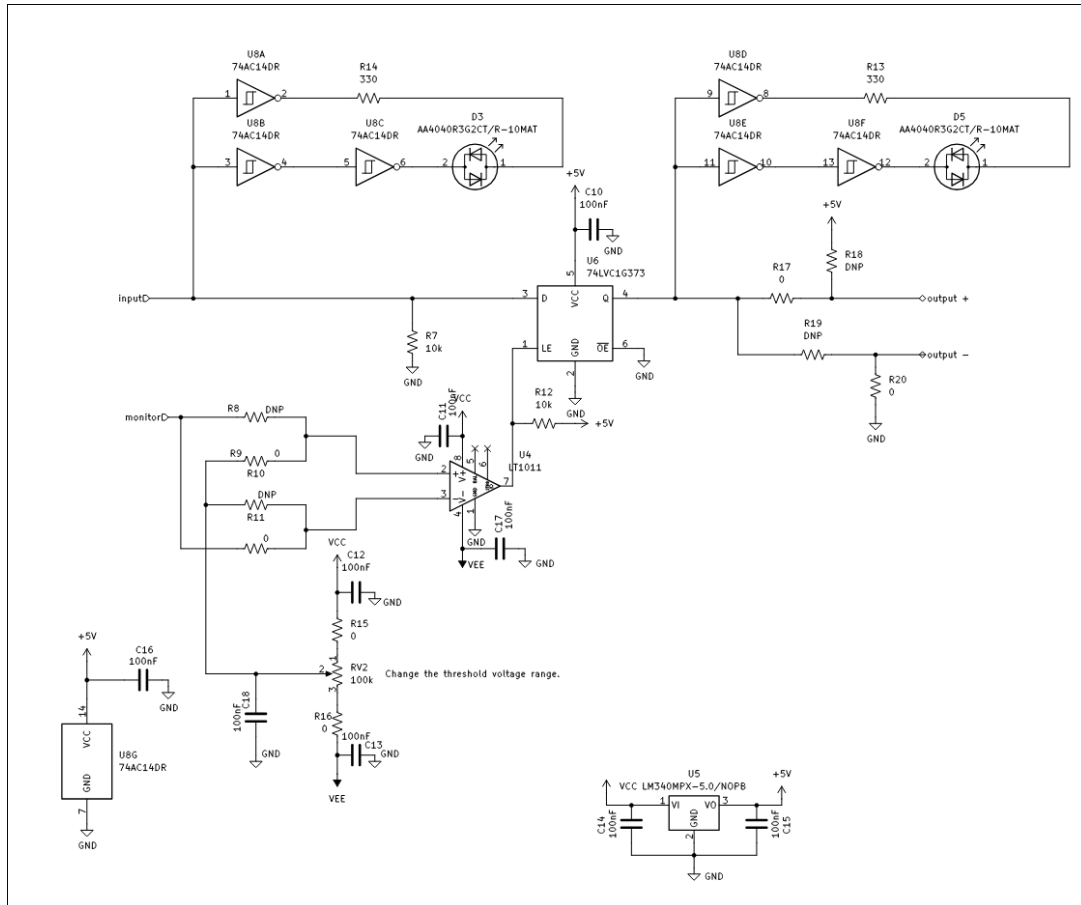


Figure 9: Protection Circuit Schematic

5.3 Protection Circuit

5.3.1 Latch Mechanism

The latch mechanism represents a critical safety feature of the MOT Coil Current Controller. This system employs a comparator circuit that continuously monitors the current flowing through the coils by comparing the Monitor voltage against a predetermined threshold voltage. This design ensures that switching between states only occurs when the current has decreased to a safe level, thereby preventing potential damage from back-EMF.

The threshold voltage can be adjusted using the potentiometer shown in Figure 4. This adjustment capability allows users to fine-tune the system’s sensitivity based on specific experimental requirements.

The protection circuit incorporates a series of inverters they process the logical signals for the latch mechanism and drive the indicator LEDs. These LEDs provide a visual indication of the system's operational state. The left LED indicates the state of the TTL input signal (low for State Green, high for State Red), while the right LED reflects the actual operational state of the MCC, which may differ from the TTL state when the current has not yet reached the safe threshold level for switching.

6 Operation

This section outlines the proper operational procedures for the MOT Coil Current Controller (MCC). Following these procedures ensures safe and reliable operation while maximizing the lifespan of the system components.

6.1 Startup and Stationary Operation Workflow

The following procedure describes the proper sequence for powering up the MCC and establishing stationary operation:

1. Provide power to the box by connecting the main power cable to a 120 VAC outlet and activating the rocker switch on the rear panel. The Power and Protection LED on the front panel should illuminate, confirming that the unit is receiving power.
2. Connect an external power supply to the MOT Coil Power Supply Input on the rear panel and set it to provide the required coil operating voltage of $\pm 15\text{V}$.
3. Apply the desired set point voltage to the Set Point input on the front panel. This voltage determines the current that will flow through the MOT coils. Refer to the Quick Reference table for the appropriate conversion ratio between set point voltage and output current.
4. Verify proper operation by monitoring the output voltage on the Monitor port, which provides feedback on the actual current flowing through the coils.

6.2 Switching Operation Workflow

Once the system is operating in stationary mode, the following procedure enables switching between State Green and State Red:

1. To request a state change, apply the appropriate TTL signal to the H-Bridge TTL input on the front panel:
 - For State Green, provide a low (0V) TTL signal
 - For State Red, provide a high (5V) TTL signal
2. When a TTL signal is applied, the left Protection LED will immediately change to indicate recognition of the TTL state request.
3. The system will not switch states immediately. The protection circuit first verifies that the current through the coils has decreased below the threshold value to prevent damage from back-EMF.
4. Once the current falls below the threshold, the right Protection LED will change color, indicating that the MCC has successfully switched to the requested state. This may take several milliseconds depending on the current level and the decay time.
5. If the right LED does not change after a reasonable time period, check the Monitor output to verify if the current is too high to allow switching. In some cases, adjusting the threshold potentiometer may be necessary (refer to Section 5.3.1).

6.3 Shutdown Workflow

To safely power down the MCC, follow these steps in sequence:

1. Reduce the Set Point voltage to zero, allowing the current through the coils to decay naturally.
2. Turn off the external Coil Operating Voltage power supply once the current has decreased to zero.
3. Turn off the main power to the MCC by flipping the rocker switch on the rear panel to the OFF position.
4. For extended periods of non-operation, it is recommended to disconnect the main power cable from the wall outlet.

6.4 Safety and Troubleshooting

When operating the MCC, observe the following safety precautions:

- Never exceed the maximum specified MOT Coil Supply voltage of $\pm 15V$.
- Do not attempt to switching by rapidly toggling the TTL input when the current is below the threshold, as this can cause excessive stress on the relay components.
- Monitor the protection LED's during initial operation to ensure all protections are functioning as expected.
- If the MCC does not power on, check the status of the fuse inside the Main Power Supply brick. Ensure $\pm 24V$ is being delivered to the Control Board.
- If there is Noise or switching capabilities are not functioning as expected, inspect the monitor wire connection between the H-bridge and the Control Board sense Resistor.
- If the cooling fan malfunctions during high operational load for any reason, the relays could be damaged. Ensure proper ventilation around the unit. Refer to the BIAS Coil Controller manual for detailed cooling considerations.

7 Appendices

7.1 Rise and Fall Time Scope Measurements

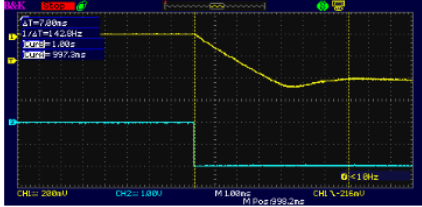
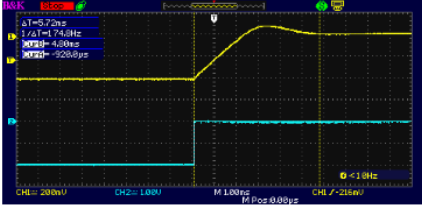
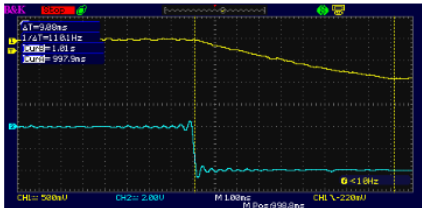
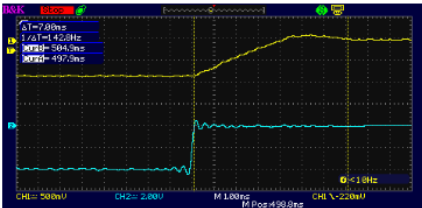
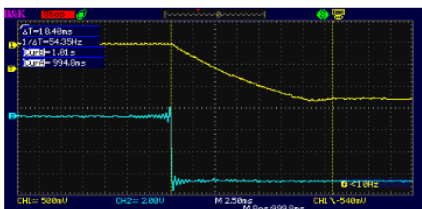
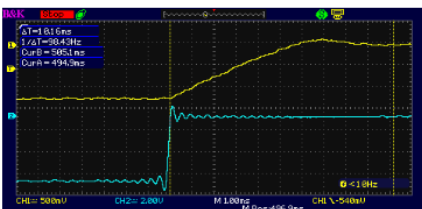
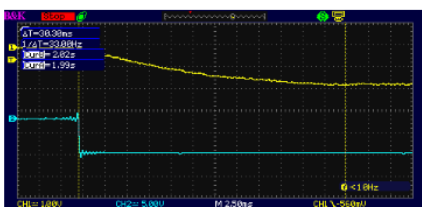
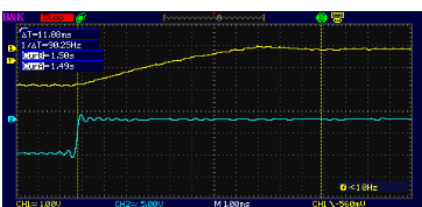
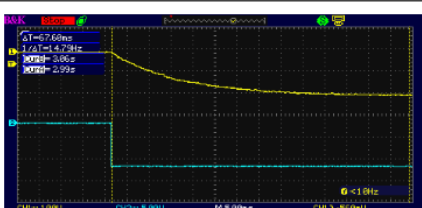
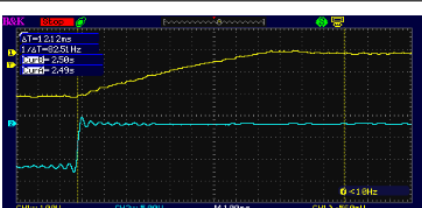
Set Point	Rise Time	Fall Time
-2V	 Full: 7.00 ms 1/e: 3.00 ms	 Full: 5.72 ms 1/e: 2.40 ms
-4V	 Full: 9.00 ms 1/e: 5.50 ms	 Full: 7.00 ms 1/e: 3.40 ms
-6V	 Full: 18.40 ms 1/e: 8.90 ms	 Full: 10.1 ms 1/e: 4.50 ms
-8V	 Full: 30.30 ms 1/e: 15.00 ms	 Full: 11.08 ms 1/e: 6.20 ms
-10V	 Full: 67.22 ms 1/e: 22.50 ms	 Full: 12.12 ms 1/e: 6.50 ms

Table 3: Rise and Fall Times for Different Set Points

7.2 Future Consideration

7.2.1 PID Loop

The components used in the PID Loop were selected based on an educated guess which has proven stable in operation. However, the PID loop has not been properly tuned as per the instructions from the BIAS Coil Current Controller Manual. A Proper tuning of the PID Loop parameters could provide slight improvements with switching time for the system.

7.2.2 Measuring Current Draw from the Negative Supply

The current draw from the negative supply of the MOT Coil Power Supply when toggling states of the MCC was not Measured during testing. For testing purposes, a 10A capable negative supply at -15V was used, which proved sufficient for all operations at the time.

If required, current draw can be measured by probing Point P; the source gate of the P-channel MOS with respect to COM/GNDD as indicated Figure 10. Then it can be compared to the Monitor Output from the front panel.

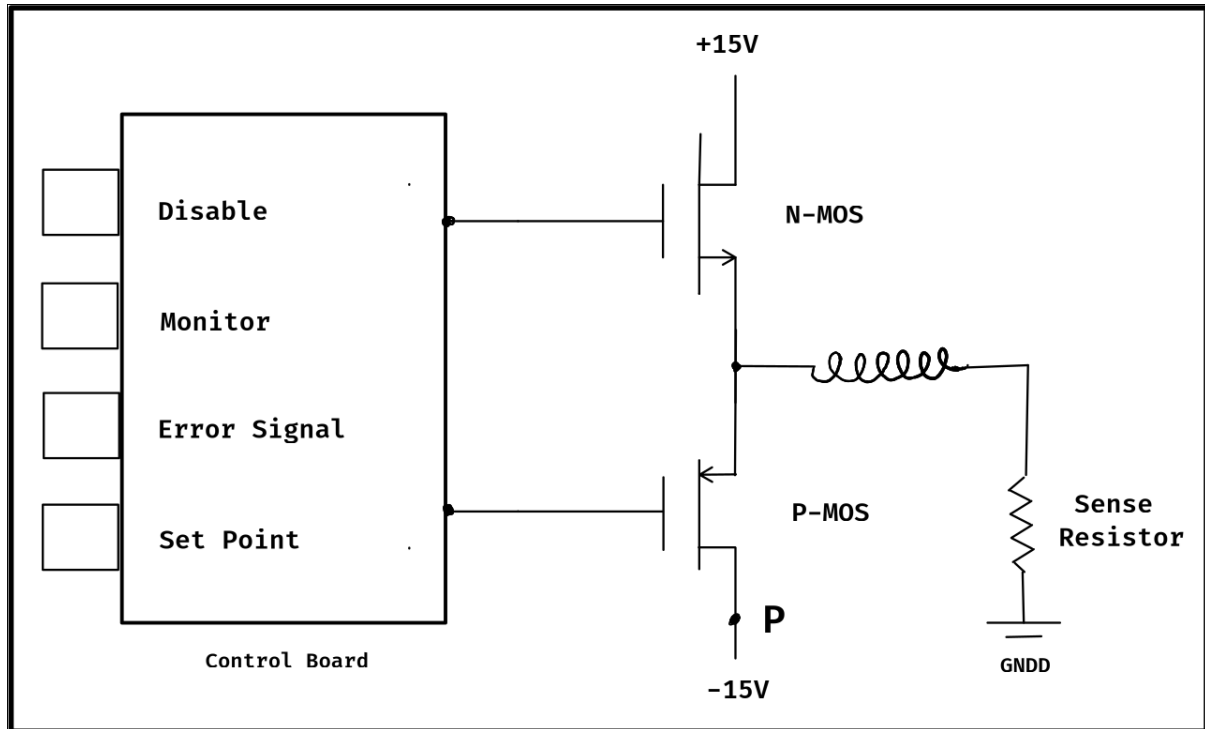


Figure 10: Current Measurement Setup for Positive and Negative Supply

It is expected that the negative supply will draw little to no current during normal operation. However, if there is uncertainty, it is recommended to use at least a 10 Amp capable power supply for the negative rail, as was done during system testing.

7.2.3 Github

- Main Repository Link: <https://github.com/JQIamo/20A-bipolar-current-source>
- H-Bridge and Relay Board Repository: https://github.com/JQIamo/20A-bipolar-current-source/tree/Single_board_and_H_bridge

The H-Bridge and Relay Repository contain all KiCad Files for the project. Additionally, it also includes other testing reports such as the MOT coil inductance measurement report and the SetPoint and Monitor Ratio calculations. It also contains videos that record the response from all 16 possible states the MCC could encounter.

7.3 Parts List

Refer to the Parts Lists from the BIAS Coil Controller Manual for the Chassis, FET-Board and Main Power input supply parts. A full Bill of Materials can also be generated using the KiCad files from the Project Repository.

Category	Part	Qty	DigiKey Part Number	Manufacturer Part Number
H-Bridge Parts	N-MOS	4	448- IAUT300N10S5N014ATMA1CT-ND	IAUT300N10S5N014ATMA1
	H-Bridge-Driver	1	HIP4080AIBZTCT-ND	HIP4080AIBZT
	Comparator	1	LT1011CS8PBF-ND	LT1011CS8PBF
	Latch	1	296-18775-1-ND	SN74LVC1G373DCKR
	Inverter	1	296-4300-1-ND	SN74AC14DR
	Voltage Regulator	1	LM340MPX-5.0/NOPBCT-ND	LM340MPX-5.0/NOPB
	80V Varistor	1	F6901-ND	V20E50P
	Rectifier Diodes	2	1727-2283-1-ND	PMEG10020AELRX
	0.047 μ F Capacitor	2	399- C0805C473K2RECAUTOCT-ND	C0805C473K2RECAUTO
	0.1 μ F Capacitor	11	311-1140-1-ND	CC0805KRX7R9BB104
	22 μ F Capacitor	4	445-14428-1-ND	C2012X5R1V226M125AC
	0 Ohms Resistor	6	A106066CT-ND	CRG0805ZR
	220 Ohms Resistor	1	A129741CT-ND	CRGCQ0805F220R
	330 Ohms Resistor	2	A129743CT-ND	CRGCQ0805F330R
	10k Ohms Resistor	5	A129761CT-ND	CRGCQ0805F10K
	100k Ohms Resistor	2	A106052CT-ND	CRG0805F100K

Category	Part	Qty	DigiKey Part Number	Manufacturer Part Number
	100k Ohms Potentiometer	1	3266Y-1-104LF-ND	3266Y-1-104LF
	4.7k Ohms Ferrite Bead	5	490-BLM21HE472SN1LCT-ND	BLM21HE472SN1L
	Otocoupler	1	160-1366-5-ND	LTV-817
	LED	2	754-AA4040R3G2CT/R-10MATCT-ND	AA4040R3G2CT/R-10MAT
	Light Pipes	2	350-3733-ND	51513020125F
	Screw Terminal 0x2	1	277-1721-ND	1984617
	Screw Terminal 0x3	1	277-1722-ND	1984620
	Terminal BLock 0x4	1	2453710000-ND	2453710000
	BNC Connector Jack, Female Socket	1	A32260-ND	5227161-7
Relay Board Parts	Mechanical Relays	2	2449-J115F11CH24VDCS61.5U-ND	J115F11CH24VDCS61.5U
	Rectifier Diode	1	1655-1N5400TB-ND	1N5400
	470 Ohms 10W Power Resistor	1	A137418-ND	SQPW10470RJ
	220 μ F Electrolytic Capacitors	1	493-12701-1-ND	UVP1H221MHD1TO
	Screw Terminal 0x2	1	277-1721-ND	1984617
	Terminal BLock 0x4	1	2453710000-ND	2453710000

Table 4: Parts List